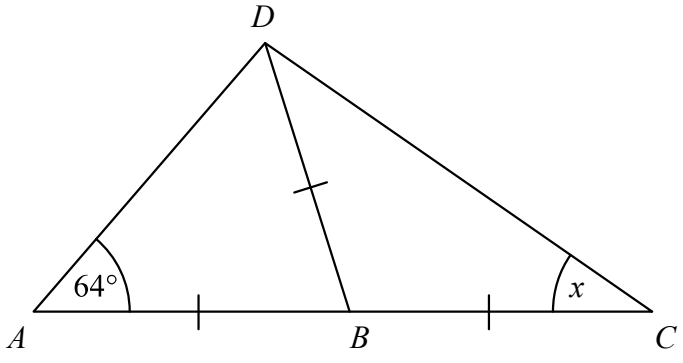
17



ABC is a straight line.
AB = *BC* = *BD*.
Angle *DAB* = 64°

Work out the size of the angle marked *x*.
Give a reason for each stage of your working.

AB = BD, so angle ADB = 64° (base angles).

Triangle ABD: angle ABD = 180° − 64° − 64° = 52°.

A, B, C are collinear: angle DBC = 180° − 52° = 128°.

BD = BC, so base angles D and C are equal.

x = (180° − 128°) ÷ 2 = 26°.

4 marks total
Equal base angles
Find ABD = 52°
Straight line = 128°
x = 26°

(Total for Question 17 is 4 marks)